Abhishek Tejrao Sose

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RESEARCH SUMMARY

Chemical engineer specializing in materials research, working across simulation and experiment to build self-driving laboratories that decide when a prediction is reliable enough to make. I carry Bayesian uncertainty quantification of force fields and machine-learned interatomic potentials (MLIPs) through molecular dynamics, Monte Carlo and DFT/AIMD to the properties that decide a material's use, and test those predictions in robotic synthesis and characterization guided by generative AI and language-model agents. Applications span metal-organic frameworks (MOFs) and their polymer composites, polymers and biopolymer composites, aerogels and alloys.

RESEARCH INTERESTS

- Bayesian uncertainty quantification of force fields and MLIPs; calibrated multiscale simulation (MD, GCMC, DFT/AIMD)
- Self-driving laboratories: generative AI, language-model agents and robotic experimentation
- MOFs and MOF-polymer composites, polymers and biopolymer composites; inverse design with explainable AI

ACADEMIC APPOINTMENTS

University of Maryland, College Park, MD 2025-present
Postdoctoral Researcher, Department of Chemical and Biomolecular Engineering | Advisor: Prof. Po-Yen Chen
Robotic experimentation guided by generative AI; language-model agents for autonomous simulation; biopolymer and aerogel composites

Georgia Institute of Technology, Atlanta, GA 2024-2025
Postdoctoral Fellow, School of Materials Science and Engineering | Advisor: Prof. Rampi Ramprasad
DFT/AIMD and MLIPs for polymers; polymer informatics; property extraction with large language models

EDUCATION

Virginia Tech, Blacksburg, VA 2018-2023
Ph.D., Chemical Engineering | Advisor: Prof. Sanket A. Deshmukh
Dissertation: *Advancing Computational Materials Design and Model Development Using Data-Driven Approaches* (December 2023)

Indian Institute of Technology (IIT) Bombay, Mumbai, India 2013-2017
B.Tech., Chemical Engineering, with a minor in Electrical Engineering

PUBLICATIONS

19 peer-reviewed papers (17 published, 2 accepted), 10 as first or co-first author; 434 citations (Google Scholar). † = equal contribution.

Journal articles

1. Development and Bayesian uncertainty quantification of transferable coarse-grained models for carbohydrates
   P Farzeen, SY Joshi, **AT Sose**, SA Deshmukh; *Journal of Chemical Theory and Computation* **2026**
2. Thermal-shock catalysts: principles, materials design, and catalytic performance
   T Li, F Bu, **AT Sose**, Z Cakir, K Zeng, AH Brozena, H Yang, C Wang, J Greeley, PY Chen, L Hu; *Nature Reviews Materials* (accepted) **2026**
3. A robotics and machine learning integrated workflow for discovering all-natural plastic substitutes
   Y Li†, **AT Sose**†, T Chen, PY Chen; *Nature Protocols* (in press) **2026**
4. Data-driven design of Cantor-type FeNiCoCrMn alloys: integrating physics-based models, explainable AI, and experiment
   F Wang, A Iwanicki, **AT Sose**, L Pressley, T McQueen, SA Deshmukh; *Digital Discovery* 5, 4093-4109 **2026**
5. Development and Bayesian uncertainty quantification of coarse-grained models of metals based on embedded atom method potentials
   AT Sose, T Gustke, F Wang, K Bejagam, A Savara, SA Deshmukh; *Journal of Chemical Theory and Computation* 21, 12481-12496 **2025**
6. Machine learning for green solvents: assessment, selection and substitution
   R Datta, J Nistane, **AT Sose**, H Sahu, R Ramprasad; *Advanced Science* e16851 **2025**
7. Photoactivated multivariate metal-organic frameworks for on-demand drug release: the role of host-guest interactions
   HD Cornell†, **AT Sose**†, S Ilic, S Chinnabattigala, N Lidman, C Oldmixon, X Yang, SA Deshmukh, AJ Morris;
   Journal of the American Chemical Society 147, 7423-7432 **2025**
8. Experimentally validated inverse design of FeNiCrCoCu MPEAs and unlocking key insights with explainable AI
   F Wang, A Iwanicki, **AT Sose**, L Pressley, T McQueen, SA Deshmukh; *npj Computational Materials* 11, 124 **2025**
9. AI-driven design of fluorine-free polymers for sustainable and high-performance anion exchange membranes
   W Schertzer, S Shukla, **AT Sose**, R Rafiq, MA Otmi, J Sampath, RP Lively, R Ramprasad; *Journal of Materials Informatics* 5 **2025**
10. Evaluation of sampling algorithms used for Bayesian uncertainty quantification of molecular dynamics force fields
    AT Sose, T Gustke, F Wang, G Anand, S Pasupuleti, A Savara, SA Deshmukh;
    Journal of Chemical Theory and Computation 20, 5732-5742 **2024**

11. Data driven discovery of MOFs for hydrogen gas adsorption
S Singh†, **AT Sose**†, F Wang, SA Deshmukh; *Journal of Chemical Theory and Computation* 19, 6686-6703 **2023**
12. A review of recent advances and applications of machine learning in tribology
AT Sose, SY Joshi, LK Kunche, F Wang, SA Deshmukh; *Physical Chemistry Chemical Physics* 25, 4408-4443 **2023**
13. Investigation of structure and dynamics of water confined between hybrid layered materials of graphene, boron nitride, and molybdenum disulfide
AT Sose, E Mohammadi, F Wang, SA Deshmukh; *Journal of Materials Science* 57, 10517-10534 **2022**
14. Determination of accurate interaction parameters between the molybdenum disulfide and water to investigate their interfacial properties
AT Sose, E Mohammadi, PF Achari, SA Deshmukh; *Journal of Physical Chemistry C* 126, 2013-2022 **2022**
15. Dual-force zone nonequilibrium molecular dynamics simulations on nanoporous metal-organic framework membranes for separation of H_2/CH_4 mixtures
F Wang, **AT Sose**, S Singh, SA Deshmukh; *ACS Applied Nano Materials* 5, 4048-4061 **2022**
16. Modelling drug adsorption in metal-organic frameworks: the role of solvent
AT Sose†, HD Cornell†, BJ Gibbons, AA Burris, AJ Morris, SA Deshmukh; *RSC Advances* 11, 17064-17071 **2021**
17. Low cost and lithography-free stamp fabrication for microcontact printing
A Khadpekar, M Khan, **AT Sose**, A Majumder; *Scientific Reports* 9, 1024 **2019**

Manuscripts submitted and in preparation

1. AeroGEM: a predictive-generative framework for composite aerogels validated with robotic experimentation
AT Sose†, T Chen†, KJ Barvenik†, A Davoodalhosseini, HC Whitley, H Yang, TC Chung, M Yu, E Tubaldi, PY Chen; *Nature Machine Intelligence* (Submitted)
2. MolOrchestra: an LLM-orchestrated, ten-agent platform for autonomous molecular simulation
AT Sose, PY Chen (In prep)

Other publications (undergraduate)

1. Oil industry - analysis, effects and removal of heavy metals
AT Sose, SJ Kulkarni, MT Sose; *International Journal of Engineering Sciences & Research Technology* 6, 254-257 **2017**
2. Characterization of sludge and design of activated sludge process for sewage treatment
SJ Kulkarni, **AT Sose**, MT Sose; *IOSR Journal of Environmental Science, Toxicology and Food Technology* 11, 20-25 **2017**

GRANTS AND COMPUTING ALLOCATIONS

1. Wrote the allocation proposals for ∼3 million NERSC core-hours over three years (PI: Prof. S. A. Deshmukh) and 1.2 million ACCESS core-hours (PI: Prof. R. Ramprasad); used more than 12 million core-hours at NERSC, the San Diego Supercomputer Center, Virginia Tech and Georgia Tech.
2. Generated preliminary data and supported proposal development for the NSF CAREER award of Prof. S. A. Deshmukh; the project's first paper appeared in the *Journal of Chemical Theory and Computation*.

AWARDS AND HONORS

1. Outstanding Student Paper Award, Computational and Theoretical Chemistry, *RSC Advances* (RSC) 2022
2. 2nd place, best talk, Carbon Nanomaterials session, AIChE Annual Meeting 2022
3. 3rd place, best talk, Inorganic Materials session, AIChE Annual Meeting 2022
4. Best Poster Award, Macromolecules Innovation Institute (MII) Technical Conference and Review, Virginia Tech 2022
5. Graduate Travel Award, Graduate and Professional Student Senate, Virginia Tech (AIChE Annual Meeting) 2022

PRESENTATIONS

Invited

1. Modelling drug adsorption in metal-organic frameworks: the role of solvent (invited webinar talk, *RSC Advances* Outstanding Student Paper Award) 2022
2. Hands-on workshop on PEUQSE, software for parameter estimation and uncertainty quantification (invited instructor, with A Savara); ACS Fall Meeting, Chicago 2022

Contributed

1. Toward confidence-gated materials discovery: Bayesian uncertainty quantification for simulation-driven design of novel materials (talk, Faculty Candidates in CoMSEF/Area 1a session); AIChE Annual Meeting, Minneapolis 2026 (upcoming)
2. Toward autonomous materials discovery through AI agents, Bayesian uncertainty quantification, machine learning, and multi-scale molecular simulations (poster, Meet the Faculty and Post-Doc Candidates session); AIChE Annual Meeting, Minneapolis 2026 (upcoming)

3. Autonomous multi-agent molecular dynamics for closed-loop discovery of sustainable biopolymer-clay nanocomposites (flash talk); 3rd Postdoctoral Research Symposium, University of Maryland, College Park 2026
4. Advancing computational materials design by leveraging data-driven approaches integrated with multiscale modeling (poster); Artificial Intelligence for Materials Science (AIMS) 2026 workshop, National Institute of Standards and Technology (NIST), Gaithersburg, MD June 2026
5. Advancing computational materials design by leveraging data-driven approaches integrated with multiscale modeling (poster, Meet the Faculty Candidates session); AIChE Annual Meeting, Boston 2025
6. Development of coarse-grained embedded atom method (EAM) potentials for FCC metals using machine learning and Bayesian uncertainty quantification (oral); AIChE Annual Meeting, Phoenix 2022
7. Predicting structure of confined water between hybrid materials using convolutional neural networks (oral); AT Sose, F Wang, SA Deshmukh; AIChE Annual Meeting, Phoenix 2022
8. Accelerated discovery of next-generation hybrid materials powered by computational materials and data science (poster, Meet the Faculty Candidates session); AIChE Annual Meeting, Phoenix 2022
9. Computational framework to develop coarse-grained EAM potentials for metals and their Bayesian uncertainty quantification (oral); AT Sose, T Gustke, F Wang, A Savara, SA Deshmukh; ACS Fall Meeting, Chicago 2022
10. Design of functionalized MOFs with enhanced CO_2 adsorption using deep-learning-assisted optimization algorithms (oral); AT Sose, F Wang, SK Singh, KK Bejagam, SA Deshmukh; ACS Fall Meeting, Chicago 2022
11. Structure and dynamics of water confined between hybrid layered materials of graphene, boron nitride and molybdenum disulfide (oral); 12th ChEGSA Symposium, Virginia Tech 2022
12. Molecular-level interactions between polymers and functionalized metal-organic frameworks (oral); MRS Fall Meeting 2021
13. Computational study of MOF-polymer compatibility in mixed matrix membranes (oral); AIChE Annual Meeting, virtual 2020
14. Molecular mechanism of gas separation through polymer-MOF mixed matrix membranes (poster); AIChE Annual Meeting, virtual 2020
15. Development of non-bonded parameters between molybdenum disulfide and water (oral); MRS Fall Meeting, virtual 2020

TEACHING AND MENTORING

1. Course development: helped Prof. Sanket A. Deshmukh build a new semester-long course on AI for simulations, Department of Chemical Engineering, Virginia Tech, Spring 2022
2. Teaching assistant, Unit Operations Laboratory (Summer 2021) and Modeling and Simulations (2020-2021 academic year), Department of Chemical Engineering, Virginia Tech
3. Invited instructor, hands-on workshop on Bayesian parameter estimation and uncertainty quantification (PEUQSE), ACS Fall Meeting, Chicago, 2022
4. Mentored 7 Ph.D. students, 1 master's student and 3 undergraduate researchers:
    - Virginia Tech: 4 Ph.D. students and 2 undergraduate researchers (Bayesian uncertainty quantification, force-field development)
    - Georgia Tech: 2 Ph.D. students and 1 undergraduate researcher (multi-task machine learning for Li-ion conductivity)
    - University of Maryland: 1 Ph.D. student and 1 master's student

SERVICE

1. Peer reviewer: *Advanced Functional Materials*, *npj Computational Materials*, *Physical Chemistry Chemical Physics*
2. Outreach, Virginia Tech: volunteered with the Center for the Enhancement of Engineering Diversity (CEED), and conducted orientations that introduced visiting high-school students to chemical engineering

RESEARCH EXPERIENCE

Postdoctoral research, Prof. Po-Yen Chen, University of Maryland 2025-present

1. *Robotic and machine-learning discovery of all-natural plastic substitutes*
    - Co-developed a workflow that couples robotic formulation and testing with machine learning (*Nature Protocols*, in press, co-first author); the work was featured by CBS News, ABC7, WTOP and FOX Baltimore.
    - Run automated formulation and testing with an Opentrons OT-2 liquid handler, a UR5e robot arm and an Instron mechanical tester.
2. *AeroGEM: predictive and generative design of composite aerogels*
    - Defined the design space of montmorillonite, carbon-nanofiber, SiO_2-nanofiber and carboxymethylcellulose aerogels with a support vector regressor.
    - Built a formulation-conditioned generative model (U-Net backbone) whose generated microstructures reproduce the pore statistics of robot-made samples' SEM images (r = 0.80-0.82, n = 277); used finite-element models for thermal, acoustic and mechanical properties and for multi-property inverse design.

3. *MolOrchestra: language-model agents for autonomous molecular simulation*
    - Building an LLM-orchestrated, ten-agent platform that plans, runs and analyzes molecular dynamics, applied to closed-loop discovery of biopolymer-clay nanocomposites.

Postdoctoral research, Prof. Rampi Ramprasad, Georgia Tech 2024-2025

1. *Polymers for Li-ion conduction: MD, MLIPs and machine learning*
    - Built a Li-ion conductivity model from 6,042 MD-simulated and 170 experimental polymers with active learning; mentored an undergraduate researcher on a multi-task model joining simulation and experiment.
2. *Machine-learned interatomic potentials for polymers*
    - Trained MLIPs on AIMD from 100 to 2,500 K and benchmarked DeepMD-kit against equivariant models (Allegro, SO3LR), which were about 250× more stable and 10× more accurate in bulk polymer MD.
3. *Data extraction and property prediction with large language models*
    - Extended a published pipeline with LLaMA and GPT-4 to extract tensile strength, Young's modulus and elongation at break from about 2.4 million full-text articles, and built Polymer Genome fingerprint models on the data.
    - Built in-context-learning and fine-tuned GPT-3.5 and Llama models for green-solvent prediction (*Advanced Science* 2025).
4. *Gas transport in polymers by non-equilibrium MD*
    - Computed CO_2, CH_4, O_2 and N_2 permeability, diffusion mechanism and solubility in 14 polymers by dual-force-zone NEMD to augment experimental data for polymer design.

Doctoral research, Prof. Sanket A. Deshmukh, Virginia Tech 2018-2023

1. *Force-field development and Bayesian uncertainty quantification*
    - Developed a Bayesian uncertainty-quantification framework for MD force fields, benchmarked sampling algorithms on an all-atom gold model, and extended it to coarse-grained EAM potentials for five FCC metals.
    - Fitted MoS_2-water cross-interaction parameters by particle swarm optimization.
2. *Metal-organic frameworks: design, separations and drug delivery*
    - Developed a PSO-integrated genetic algorithm, accelerated with Gaussian-process, random-forest and CNN surrogates, to search $\sim 10^{37}$ functionalized MOFs, finding a MOF with 6× the H_2 storage of the bare framework; identified 19 amine functional groups that enhance CO_2 adsorption.
    - Simulated Xe/Kr adsorption in 7,786 CoRE MOFs with rigid and flexible frameworks: frameworks that were Kr-selective when rigid became Xe-selective when flexible. Screened ∼137,000 hypothetical MOFs by GCMC; leading building blocks went to collaborators at Pacific Northwest National Laboratory for synthesis.
    - Modeled light-responsive MOFs for anticancer drug release with an experimental group (A. J. Morris), and MOF-polymer interfaces and mixed matrix membranes for H_2/CH_4 separation by steered and non-equilibrium MD.
3. *Water in layered materials and deep learning*
    - Simulated water confined between graphene, boron nitride and MoS_2, and trained 2D and 3D CNNs to classify and predict confined-water structure.

Undergraduate research 2014-2017

1. 3D-printed PLLA and PCL tissue scaffolds: pore morphology and cell growth; Loughborough University, UK (Prof. D. Das) 2016
2. Led plant operations of Biosynth, a student-run 250 L biodiesel pilot plant, and mentored 30 sophomores; IIT Bombay (Prof. S. M. Mahajani) 2014-2016
3. Process design of an integrated gasification combined-cycle plant in Aspen Plus; IIT Bombay (Profs. Y. Shastri and V. M. Naik) 2017

SKILLS

Simulation:	LAMMPS | GROMACS | NAMD | VASP | Gaussian | RASPA | CHARMM-GUI
MLIPs and ML:	DeepMD-kit | Allegro | SO3LR | PyTorch | JAX | TensorFlow/Keras | Bayesian inference (PEUQSE, Zeus)
LLMs and agents:	LLaMA and GPT models | fine-tuning | in-context learning | retrieval-augmented generation | multi-agent workflows
Lab automation:	Opentrons OT-2 liquid handler | UR5e robot arm | Instron mechanical testing | SEM
Programming:	Python | C++ | Fortran | Bash | Tcl | SQL
Methods:	GCMC | dual-force-zone NEMD | steered MD and free energies | AIMD | finite-element modeling

MEDIA COVERAGE

1. Featured researcher, AI-driven biodegradable shellfish-based packaging, **FOX Baltimore** 2026
2. Featured researcher, "UMD lab creates environmentally-friendly plastic products to cut down pollution", **CBS News** 2026
3. Featured researcher, "UMD team nears breakthrough in fight against plastic pollution", **ABC7 News** 2026
4. Featured researcher, "AI is helping these Terps create a biodegradable plastic cling wrap", **WTOP News** 2026